



PURPOSE-BUILT FOR POST-PURCHASE RISK

Stop return fraud before it becomes refund loss.

Score every return request, explain the decision, and block risky refunds before cash leaves the system.

Built for the fraud surface checkout tools miss: empty-box claims, wardrobing, item swaps, fraud rings, and suspicious return narratives.

0-100

RISK SCORE

3

DECISION BANDS

Explain

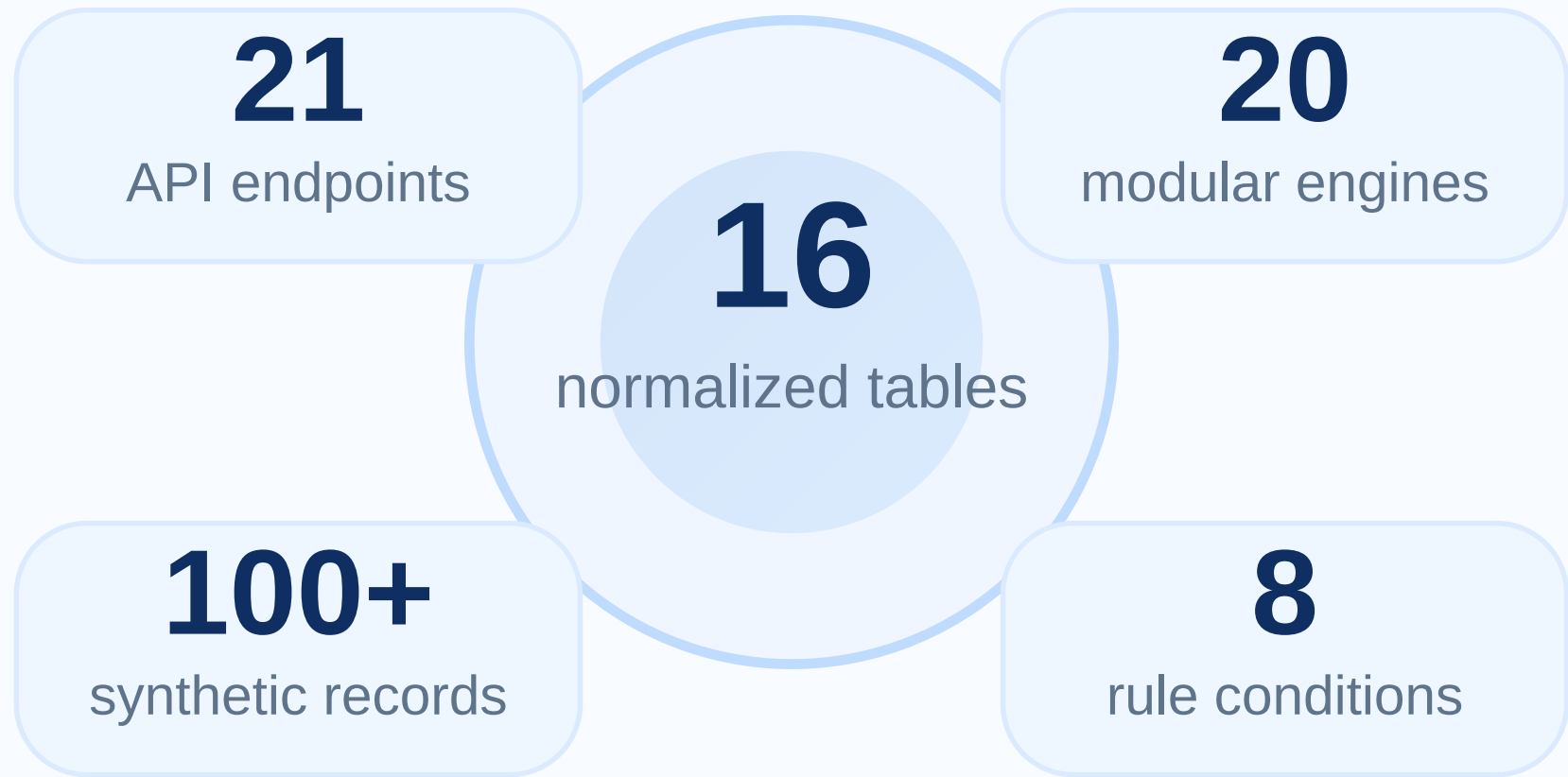
REASON TRACE

Live

API SCORING

SYSTEM FOOTPRINT

REAL REPO COUNTS



5
MODEL FAMILIES

3
DECISION BANDS

28+
CSV ALIASES

9
FRAUD PATTERNS



THE PROBLEM

Legacy fraud stacks miss the returns layer.

DIMENSION	PAYMENT FRAUD	RETURN FRAUD
Detection point	Checkout	Post-fulfillment
Signals	~50 features	200+ features across order, text, graph, images
Attack surface	Card details	Physical goods, labels, weight, chat, photos

Existing coverage

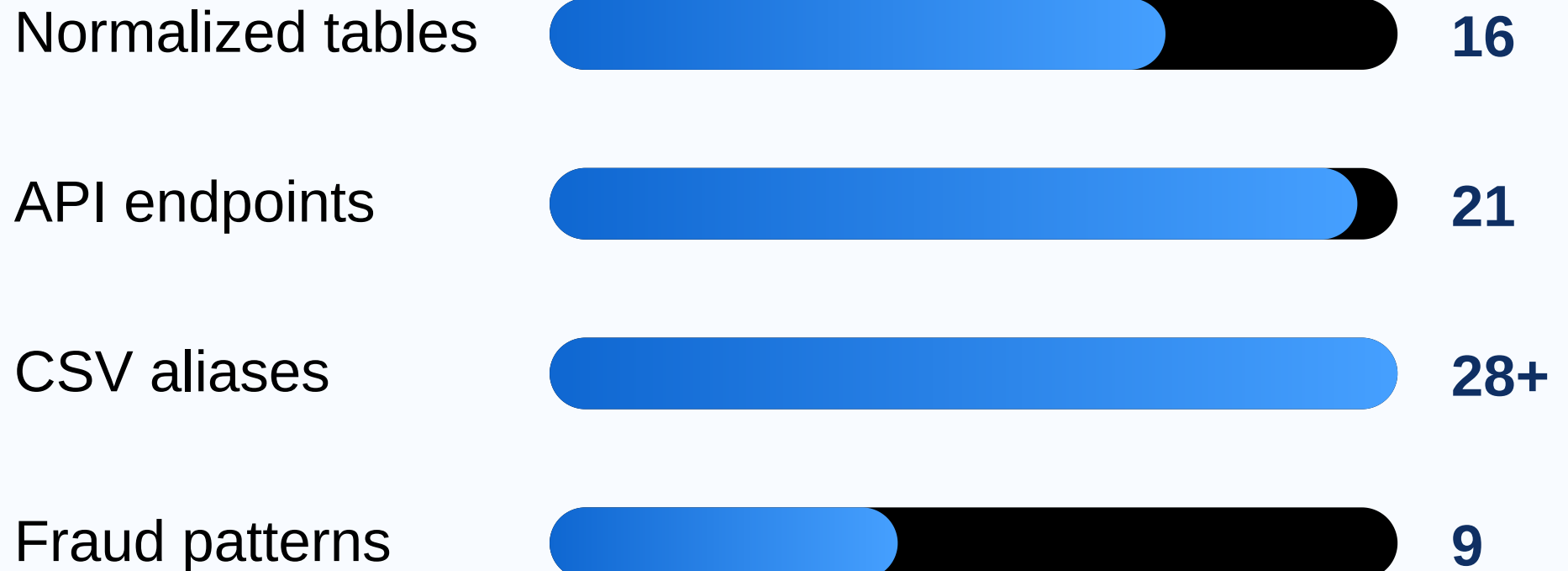
Strong

Near zero

SURFACE AREA

WHAT THE SYSTEM REALLY SEES

Counts from the codebase and docs





PRODUCT FEATURES

**One workflow
from scoring to
analyst action.**

DECISION MIX

LIVE SCORING PATH

Rules and supervised ML drive the live score

35% rules

65% supervised ML

final_score = rule-led supervised scorer

NLP, graph, and anomaly signals remain available for explanation.

Rules
35%

Supervised ML
65%

Fallback
HEURISTIC

Explain
ALWAYS ON



ALGORITHM

Rules and supervised ML drive one defensible decision.



How the model works



ACCURACY

**Accuracy comes
from layered
validation, not
black-box claims.**

The repository documents scoring and validation mechanics, but it does not publish audited benchmark percentages. That keeps the pitch credible and print-safe.

EVIDENCE DECK

VALIDATION INPUTS

Real counts from the repo, not benchmark claims

100+
records

9

3

7

synthetic seed fraud patterns tracked metrics seed rules



ADOPT FAST

Score, explain, review, and learn in one control loop.

ReturnShield AI turns returns into a measurable control point for merchants who need to stop refund loss without slowing legitimate customers.

1M+ row generator, 50k unique customers, 28+ aliases, 21 endpoints

1M+ synthetic rows

50k unique customers

28+ CSV aliases

21 API endpoints

